

$$t_{n+1} = t_n + h, \quad \mathbf{x}_{n+1} = \mathbf{x}_n + h\mathbf{f}(t_n, \mathbf{x}_n)$$

where $\mathbf{x} = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$ and $\mathbf{x}'(t) = \mathbf{f}(t, \mathbf{x}) = \begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix}$.

We are given the following conditions:

$$\begin{aligned} x' &= -x + y - (1 + t^3) e^{-t}, & x(0) &= 0 \\ y' &= -x - y - (t - 3t^2) e^{-t}, & y(0) &= 1 \\ x(t) &= e^{-t} (\sin(t) - t), & y(t) &= e^{-t} (\cos(t) + t^3). \end{aligned}$$

Our goal is to use **Euler's Method** to approximate values $x(1)$ and $y(1)$ using step size $h = 0.05$ to five decimals.

First, we must establish the Fundamental Theorem of Calculus:

$$y(t_n) - y(t_0) = \int_{t_0}^{t_n} y'(t, y(t)) dt.$$

Rearranging this equation and letting the total integral be the sum of individual partitions of that area, we can state:

$$y(t_n) = \sum_{i=0}^{n-1} \int_{t_i}^{t_{i+1}} y'(t, y(t)) dt + y(t_0).$$

We can't directly integrate y' since we don't know the value for $y(t)$, so our next best option is to approximate the integral. Instead of worrying about the curvature of the function, we can simply start at t_i and extend a rectangle toward t_{i+1} :

$$\int_{t_i}^{t_{i+1}} y'(t, y(t)) dt \approx y'(t_i, y(t_i)) \cdot (t_{i+1} - t_i).$$

Substituting this back into our equation yields:

$$y(t_n) = \sum_{i=0}^{n-1} [y'(t_i, y(t_i)) \cdot (t_{i+1} - t_i)] + y(t_0).$$

We know that $(t_{i+1} - t_i)$ is equal to the change in t ; therefore being equal to h . Allowing us to say:

$$y(t_n) = h \sum_{i=0}^{n-1} [y'(t_i, y(t_i))] + y(t_0).$$

Approximating t_i by extrapolating forward in time using our change in t , we can simplify this further and state:

$$y(t_n) = h \sum_{i=0}^{n-1} [y'(t_0 + i \cdot h, y(t_0 + i \cdot h))] + y(t_0), \quad \text{since } t_i = t_0 + i \cdot h.$$

Subbing in our values for h and t_0 yields:

$$y(t_n) = 0.05 \sum_{i=0}^{n-1} [y'(0.05i, y(0.05i))] + 1.$$

And because $h = 0.05$, the following must be true:

$$\begin{aligned} h &= \frac{t_n - t_0}{n} \\ \implies n &= \frac{t_n - t_0}{h} \\ &= \frac{1 - 0}{0.05} = 20. \end{aligned}$$

This gives us our final equation defining $y(1)$:

$$y(1) = 0.05 \sum_{i=0}^{19} y'(0.05i, y(0.05i)) + 1.$$

We can apply the same logic to x by integrating its derivative $x'(t)$ from $t_0 = 0$ to t_n as well giving us the equation:

$$x(t_n) = 0.05 \sum_{i=0}^{n-1} [x'(0.05i, x(0.05i))] + 0.$$

Having these equations is great, but the main problem with them is that they cannot be computed directly, as they depend on themselves (implicit). We are forced to solve at each index numerically to achieve each subsequent value (breaking it down into multiple recursive equations). For the equation defining $y(t_n)$, for the initial step, $n = 1$, we achieve: $y(t_1) = 0.05 \sum_{i=0}^0 y'(0.05i, y(0.05i)) + 1$. Because at $i = 0$, all values are their initial values, we get:

$$\begin{aligned} y(t_1) &= 0.05 \sum_{i=0}^0 y'(0, y(0)) + 1 \\ &= 0.05 [y'(0, 1)] + 1 \\ &= -0.05 + 1 \\ &= 0.95. \end{aligned}$$

This is now our initial approximation for y_1 where $t_1 = t_0 + h = 0 + 0.05 \implies y(t_1) = y(0.05)$. Doing this again up to $i = 1$ would yield:

$$\begin{aligned} y(t_2) &= 0.05 \sum_{i=0}^1 [y'(0.05i, y(0.05i))] + 1 \\ &= 0.05 [y'(0, 1) + y'(0.05, 0.95)] + 1 \\ &= 0.05 [(-1) + (-0.99043)] + 1 \\ &= 0.05 [-1.99043] + 1 \\ &= -0.09952 + 1 \\ &\approx 0.90048. \end{aligned}$$

This is the recursive concept, but I won't bore you with the 18 other steps. Running this procedure across all 20 iterations for both $x(t_i)$ and $y(t_i)$ allows us to build up the entirety of each curve up to $t = 1$. Doing so results in the final computed datasets as shown in Table 1:

Table 1: Computed x_i and y_i Values

i	0	1	2	3	4	5	6	7	8	9	10
t_i	0.00	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50
x_i	0.00000	0.00000	-0.00007	-0.00033	-0.00088	-0.00179	-0.00312	-0.00490	-0.00714	-0.00986	-0.01305
y_i	1.00000	0.95000	0.90048	0.85229	0.80614	0.76260	0.72213	0.68507	0.65168	0.62213	0.59654

i	11	12	13	14	15	16	17	18	19	20
t_i	0.55	0.60	0.65	0.70	0.75	0.80	0.85	0.90	0.95	1.00
x_i	-0.01669	-0.02075	-0.02522	-0.03004	-0.03519	-0.04062	-0.04630	-0.05216	-0.05818	-0.06431
y_i	0.57495	0.55735	0.54369	0.53388	0.52781	0.52532	0.52625	0.53041	0.53760	0.54761

The values presented in Table 1 represent the final numerical approximations of $x(t_i)$ and $y(t_i)$ at each index of the summation. Taking our values at the final step where $i = 20$ and $t = 1$, we achieve our final approximations for the system:

$$\begin{aligned} x(1) &\approx -0.06431 \\ y(1) &\approx 0.54761 \end{aligned}$$